



UNIVERSAL ROBOTS

A Teradyne Robotics Company

Certificate of participation

Advanced Training

Certificate number: E202309291970810

Robot type: e-Series

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21. September 2023

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Universal Robots Academy Mexico



Curriculum

Julio Antonio Caballero Mora 21. September 2023

Module 1: Program structure and concepts

- How to create a program flow chart
- Requirements to a good robot program
- How to create a good program structure

Module 2: Basics of URScript

- What is URScript
- How to write your own script functions
- Pass parameters to functions and return values from functions
- Index a pose and/or list variable

Module 3: Pose variables

- Structure of a pose variable
- Script commands `pose_add()` and `pose_trans()`
- Differences between `pose_add()` and `pose_trans()`
- How to use the `get_actual_tcp_pose()` script command

Module 4: Coordinate systems

- Use a coordinate system as a variable
- Program relative to a coordinate system variable
- Transfer the programming to another coordinate system
- Move coordinate system variables inside of a program

Module 5: Advanced use of TCP

- Teach in the position and orientation of a TCP
- Adjust the center of gravity in the robot program
- Switch between two TCPs in a robot program

Module 6: Conveyor tracking

- Configure the conveyor tracking wizard
- Use the conveyor tracking wizard in a program
- Pick parts up from a moving conveyor

Module 7: Force control (advanced)

- Configuration of various force functions
 - Force control (frame)
 - Force control (motion)
 - Force control (point)



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